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Content companies struggle to safeguard their content even as crackers find new methods of defeating them. In the end, it's the consumer who pays for products that don't work

Content is king, and nowhere more so than in the entertainment industry. It brings in the money, reins in the audience and is thus zealously guarded by the corporate moguls of the music and movie world. Lately, technology has been the prime cause of angst for these sectors. A rise in music and motion picture piracy has been directly blamed on advances such as the rise of the Internet, the advent of efficient compression technologies and the falling prices of both CD recorders and recordable media. Experts in these industries readily blame the fall in the sales of their products on these factors. Hilary Rosen, president and chief executive officer of the Recording Industry Association of America (RIAA) has this to say about the scenario: "A large factor contributing to the decrease in overall shipments of music CDs is online piracy and CD burning. Global piracy on the physical side costs the recording industry over \$4 billion a year." The music industry was especially caught off-guard by the spread of MP3 files on the online service, Napster. Peer-to-peer services such as

Gnutella and KaZaA have had a similar impact on the movie industry—pirated copies of DVD movie discs trickle down from the Internet and appear in potential markets much before the release of the feature film in cinema halls.

To consolidate their position and market share, both these industries have taken measures to dampen the technological fallout and ensure that such scenarios do not repeat themselves. One such measure is the use of copy-protection technologies to guard their valued content against would-be pirates. Join us as we take a look at how copy protection techniques used on both CD-ROM and DVD-ROM drives are affecting the buying public and how they are implemented.

Wherever you may ROM

A CD is ubiquitous; it brings us entertainment in the form of video CDs, audio CDs and games. Content on this medium is also the most pirated. A survey suggests that sales of CD recorders have tripled since 1999, that there are approximately 29.9 million

unauthorised CD-ROM copies in US households, and that almost half of the software copied are games, with educational software coming in at a close second. The paranoia over protection of media has spawned an entire industry that aims to provide solutions to the content generators. Some of these schemes bend the standards that are laid down for the production of CD-ROMs. Some even go as far as breaking them. The issue is real enough to have caused Philips (the co-creator of the CD-ROM standard) to raise concerns that users might be buying discs that are not 'true CD-ROMs'.

Breaking the standards: The easiest means of content protection involves increasing the size of the CD medium.

Dummy files, which merely act as pointers to other files on the CD are used to do just this. This way the CD is made to look bigger than it is—greater than its standard 600 MB capacity, and any attempt to copy files from the CD to the hard disk (a process that some pirates use



According to the International Planning and Research Corporation, the losses due to software piracy in 2001 amounted to \$10.97 billion